Engin 29 Final Project

Team 101-1

Project Name: Desk Extender

Spring 2026

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Video Link:

https://drive.google.com/file/d/11FkBVWuMpfkhLuJlVebbYfmFK-Mbas_P/view?usp=sharing

Material and Process Selection Table:

The material and process selection table is attached externally to BCourses and in Appendix A.
Link to spreadsheets: Material Selection and Fit Spreadsheet

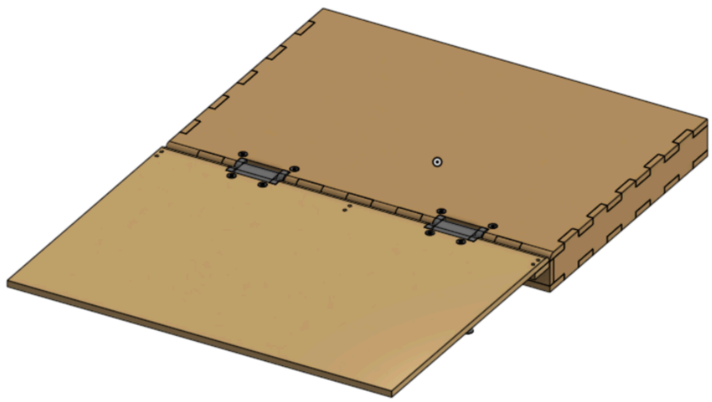
Fits Table:

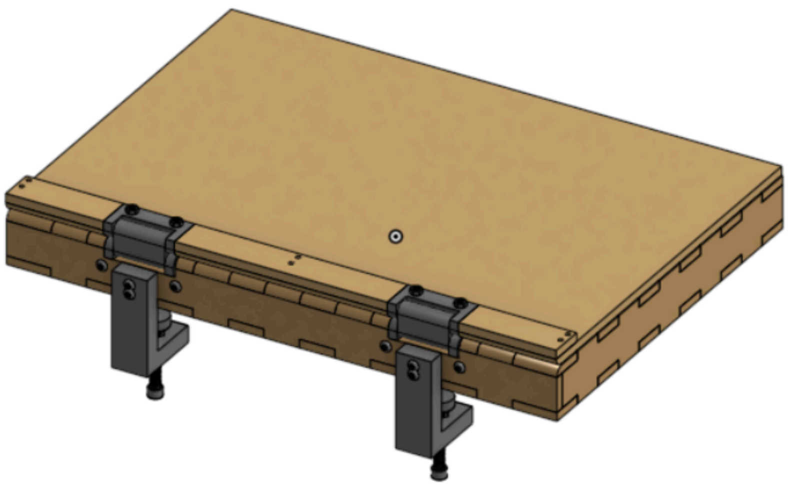
The fits table is attached externally to BCourses and in Appendix B.
Link to spreadsheets: Material Selection and Fit Spreadsheet

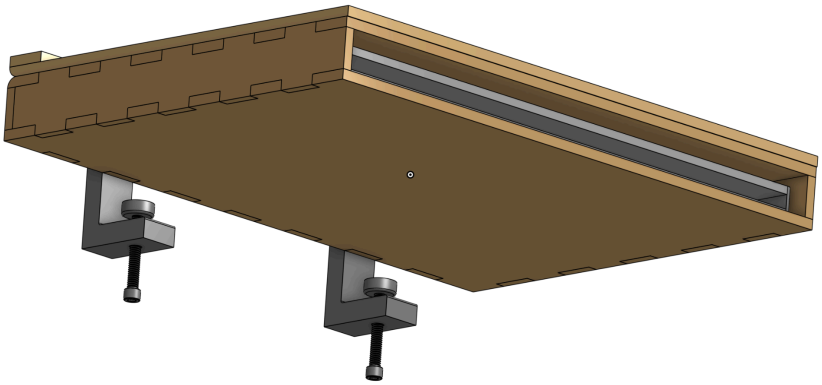
Final Photographs:

Final CAD Model:

Onshape Link:
https://cad.onshape.com/documents/d71e4eeebce875e60bad1736/w/99ba947de1130029e2c9cb7c/e/3049beb7b776da85c2f05cf0?renderMode=0&uiState=6a02b39e107f55d52a790c95







Design Calculations:

Material selection:

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--- Factor of Safety (FOS) Comparison ---
Target: Laptop load of 8 lbs on a 15-inch arm.
Material        Aluminum 6061-T6  Delrin (Acetal)  Plywood (Birch)  \
Thickness (in)
0.125                      16.05             4.59             2.29
0.250                      64.22            18.35             9.17
0.375                     144.49            41.28            20.64
0.500                     256.87            73.39            36.70
0.750                     577.95           165.13            82.56

Material        Polycarbonate
Thickness (in)
0.125                    4.13
0.250                   16.51
0.375                   37.15
0.500                   66.05
0.750                  148.62

--- Maximum Deflection (inches) Comparison ---
Material        Aluminum 6061-T6  Delrin (Acetal)  Plywood (Birch)  \
Thickness (in)
0.125                     0.1407           3.5173           0.9379
0.250                     0.0176           0.4397           0.1172
0.375                     0.0052           0.1303           0.0347
0.500                     0.0022           0.0550           0.0147
0.750                     0.0007           0.0163           0.0043

Material        Polycarbonate
Thickness (in)
0.125                  4.0197
0.250                  0.5025
0.375                  0.1489
0.500                  0.0628
0.750                  0.0186
```

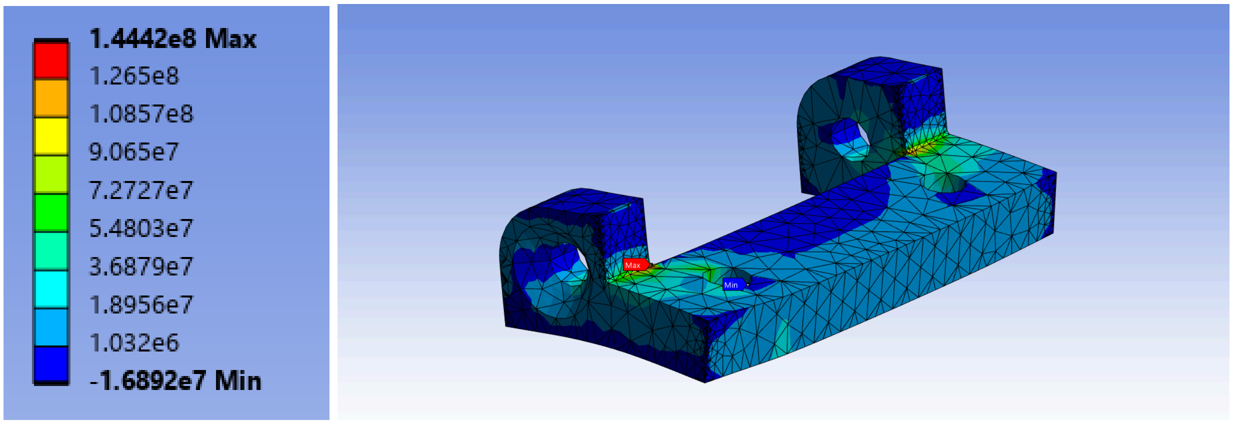
While certain materials boasted a much higher factor of safety, the deflection showed that some of those materials bent significantly, reducing the effectiveness of a cantilevered desk extender. As a result, we felt that 0.25" thick plywood was the ideal in-between material and thickness.

PLA orthotropic material property values found + interpolated from "Orthotropic mechanical properties of PLA materials fabricated by fused deposition modeling" M. Li et al.

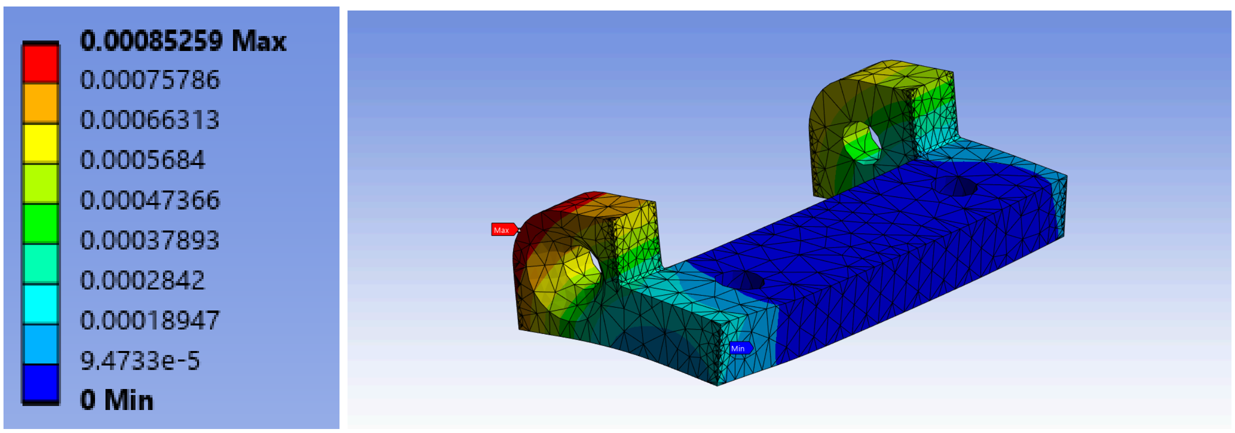
Table of Properties Row 3: Orthotropic Elasticity

	A	B	C	D	E	F	G	H	I	J
1	Temperature (C)	Young's Modulus X direction (Pa)	Young's Modulus Y direction (Pa)	Young's Modulus Z direction (Pa)	Poisson's Ratio XY	Poisson's Ratio YZ	Poisson's Ratio XZ	Shear Modulus XY (Pa)	Shear Modulus YZ (Pa)	Shear Modulus XZ (Pa)
2		2.669E+09	2.583E+09	2.208E+09	0.43	0.4	0.37	9.19E+08	8.82E+08	8.44E+08
*										

Maximum Principal Stress



Deformation



Shows a maximum stress that exceeds the failure stress for PLA. Running the simulation again with Aluminum Alloy as the material showed a stress that was well within the failure stress for Aluminum, showing that while our PLA prototype hinge would fail, a CNC'd or OTS aluminum hinge could withstand the loads in our product.

Conclusions and Prototype Reflection:

Overall, our prototype was pretty good. Our prototype was primarily made of plywood, some 3d printed components, some aluminum, and some screws and nuts. For an initial prototype, the desk extender served both purposes it was intended for. First, it opened up nicely to increase desk space without being an inconvenience for others while staying very secure to the small existing desk. Second, it provided enough space to store a laptop without damaging it or forcing it in. Although the plywood joints were made with wood glue, the strength was not bad. This is something we would likely want to change since, over time, the joint would likely weaken. We also believe that 3d printed parts are not the strongest for this purpose. Thus, creating hinges out of lightweight but strong materials like composites or using injection molding could make both the process and the strength of our product better. However, as we did not have access to some of these materials and processes, we are pretty happy with our prototype. Some future functionalities we could add would be a more compact attachment mechanism for the desk and a possible latch-like mechanism to prevent a laptop from falling out. Our pocket, which is meant for a laptop, can also be used as a storage space for pencils and notebooks if someone does not want to put a laptop in there. Our overall prototype is a good representation of our ability to design, test, and actually build a functional tool that solves real-world problems. Thank you to the course staff for supporting our project, providing us with really good advice, and an awesome semester!

Appendix A: Materials and Processes Table

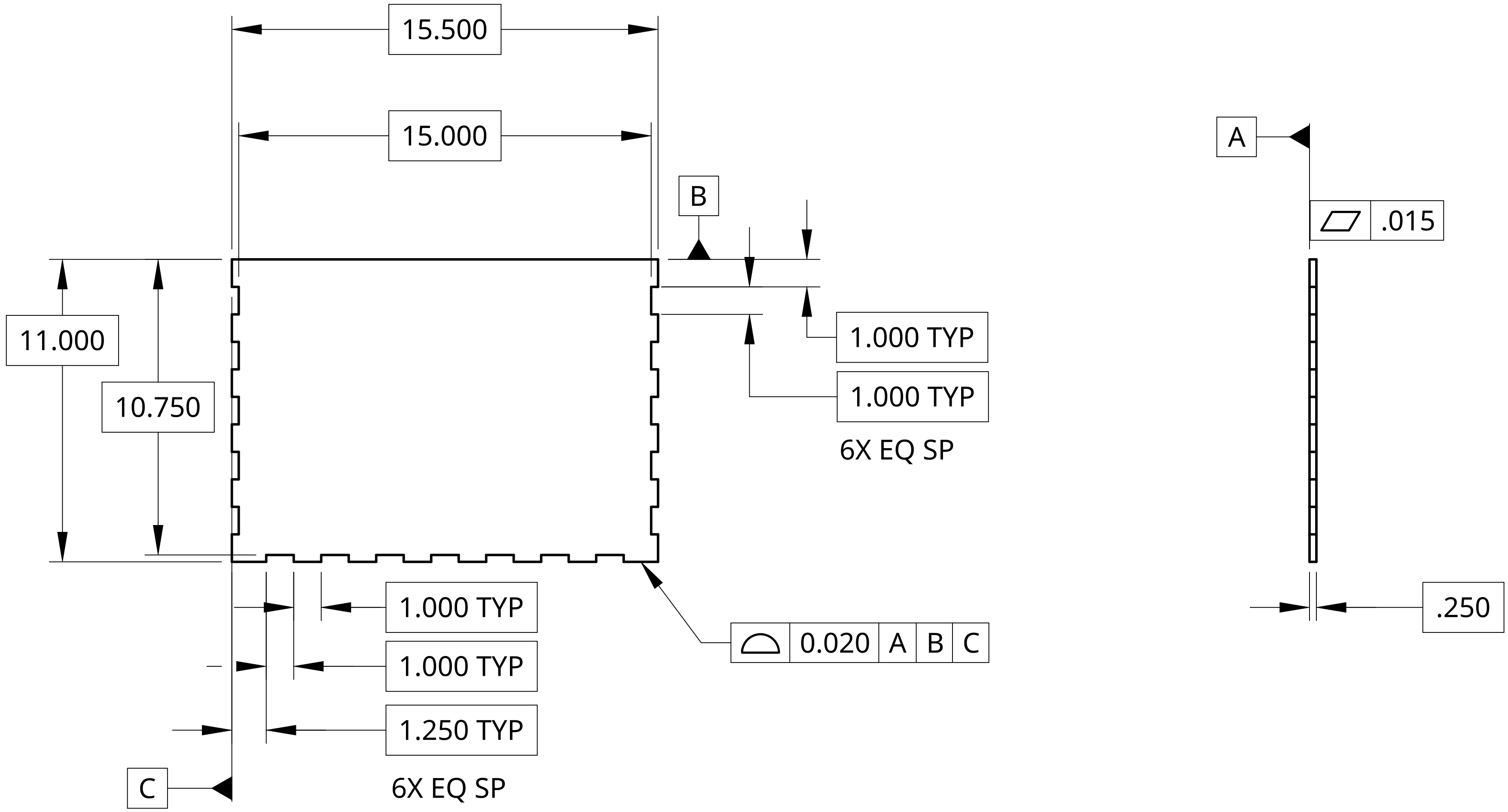
Component (or component group) #	Component name	Material candidate	Process candidate	Advantages of material/process	Potential disadvantages of this material/process combination	Selected material/process combination, and brief justification
1	Knob Mount	Aluminum	CNC	Stronger material, durable, and gives the part a sleek/professional look	More expensive material and more difficult/time-consuming to manufacture	3D printing for demo; CNC for market. For our proof of concept, the knob mount does not need extreme durability, so PLA 3D printing is faster, cheaper, and easier to modify. For a final market version, CNC aluminum would be better because it is stronger, more durable, and has a cleaner finish.
		PLA	3D printing	Fast, cheap, and easy to manufacture. Good for quick prototyping and design changes	Low durability compared to metal and may wear or break under repeated use	3D printing for demo; CNC for market. PLA is best for the prototype because it saves time and cost, but aluminum would be better for long-term use.
2	Fold out top	Wood	Laser cutter	Fast to make, clean appearance, and works well for flat pieces	Can split or break while drilling, especially near edges or thin areas	Laser cutting. Since this component is mostly flat, laser cutting is faster and gives a cleaner, sleeker design than 3D printing.
		PLA	3D printing	Requires little manual labor and can create more complex shapes	Time-consuming for a large flat part, limited by printer bed size, and may have a less attractive surface finish	Laser cutting. PLA is possible, but laser-cut wood is more efficient and better-looking for this type of flat component.
3	Wood laptop holder	Wood	Laser cutter	Fast to make, nice-looking finish, and good for consistent flat parts	Pieces may need to be glued manually, and it is harder to create complex	Laser cutting. Wood and laser cutting are best because the part is mostly flat, can be
		PLA	3D printing	Requires little manual labor and allows for more complex 3D geometry	Limited bed/print space, longer print time, and rougher or less professional appearance	Laser cutting. PLA would take longer and may not look as clean, so laser-cut wood is the better option.
	Hinge	Aluminum	CNC	Better look and feel, stronger hold, and more durable under repeated movement	Metal is expensive, and it takes more time to drill and machine the required holes	3D printing for prototype; metal/injection molding for market. For the prototype, PLA is cheaper and faster. For a final product, metal or injection molding with metal reinforcement would be better for strength and mass manufacturability.

4		PLA	3D printer	Fast to make, cheap, and easy to modify during prototyping	Not very durable and may wear down or fail with repeated hinge movement	3D printing for prototype; stronger process for market. PLA works for testing, but a stronger process would be needed for long-term use.
	Clamp Rod	Aluminum	Buy premade	Strong, durable, sleek-looking, and reliable for carrying load	More costly than plastic alternatives	Aluminum. This part carries a lot of load, so we need a strong and durable material. Buying it premade also saves manufacturing time and improves reliability. Was made for our first design was abandoned during our redesign
5		PLA	3D print	Easy to manufacture and inexpensive	Not strong enough for the load and could bend, crack, or fail	Aluminum. PLA is not a good choice because the clamp rod is load-bearing and needs higher strength.
	Clamp Pivot Bar	PLA	3D printing	Cheap and can be strong enough depending on geometry	Time-consuming to print compared to cutting a flat part	Laser cutting. Since this is a flat piece and does not bear a lot of load, laser cutting is faster and gives a better-looking result. Was made for old design was abandonded in newer version
6		Wood	Laser cutter	Fast, cheap, and gives a sleeker appearance	Weaker than metal and may not perform as well under high stress	Laser cutting. Wood is acceptable because the part is not heavily loaded, and the process is faster than 3D printing.
	Clamp Beam mount	PLA	3D printing	Cheap, fast, and easy to prototype	Weaker than aluminum and may deform under higher loads	PLA. This part does not need to bear a large amount of load, so PLA is strong enough while also being cheaper and faster to make. Were made for previous design was reworked during redsign
7		Aluminum	Laser cutting	Strong and sleek-looking	Expensive material and may be unnecessary for a lightly loaded prototype part	PLA. Aluminum would be stronger, but it adds cost without being necessary for this part.
	Hinge holder	PLA	3D printing	Fast to make, cheap, and easy to modify	Weaker than aluminum and less durable over time	PLA. Stress analysis showed that this component does not need to be very strong to support the expected load, so PLA is sufficient for the prototype.

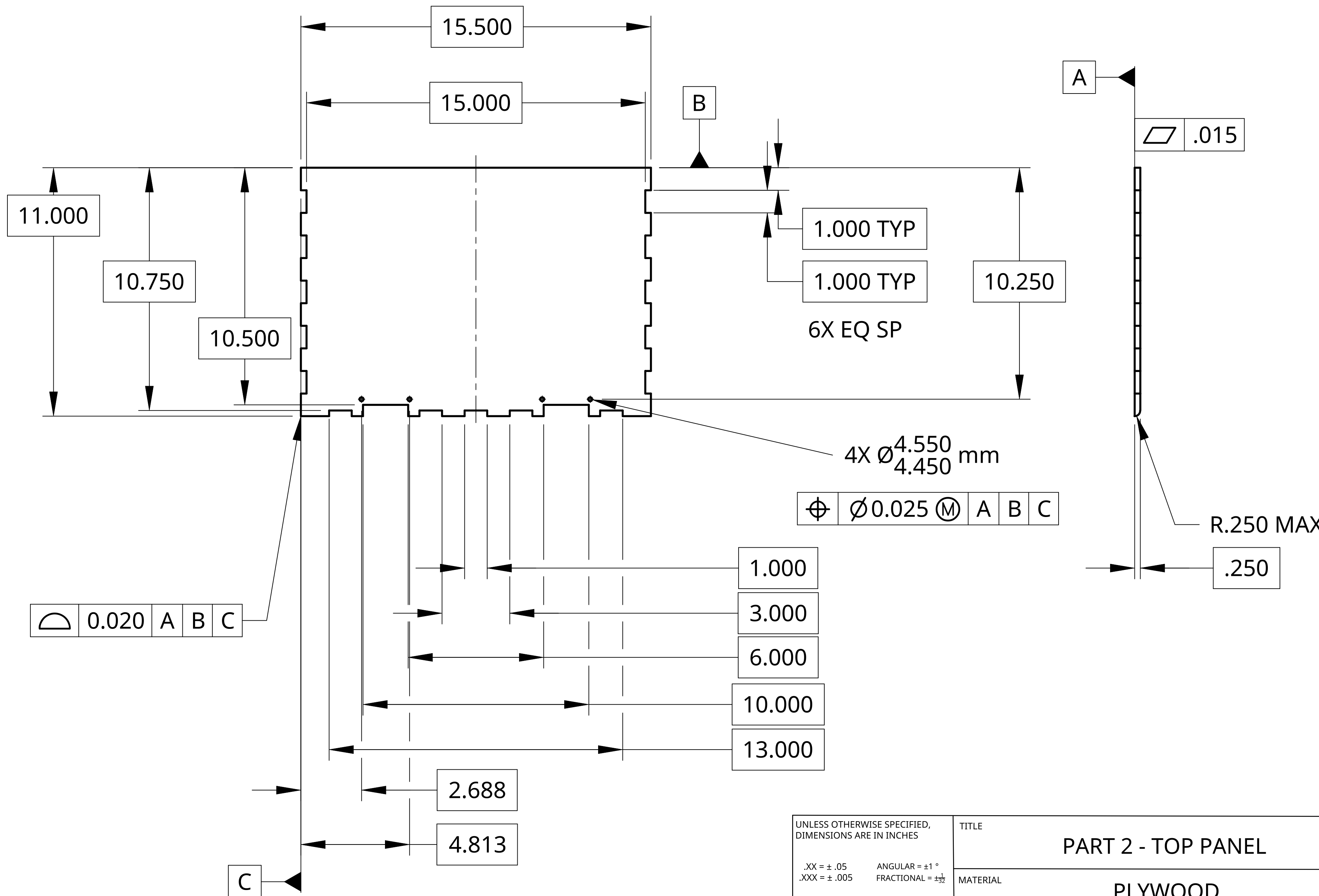
8		Aluminum	CNC	Strong, durable, and better for long-term use	Expensive and takes more time to machine	PLA. Aluminum would work, but it is unnecessary because the stress analysis showed the part does not need that much strength.
9	Hinge Standoff	Aluminum 6061	Machining	Strong, durable, and relatively easy to machine	Time-consuming, less consistent than automated processes, and may involve more human error	Machined Aluminum 6061. This part carries a lot of stress, so strength and durability are more important than speed or cost.
		PLA	3D printing	Can be made with the other small parts, cheap, and easy to prototype	Weak compared to aluminum and more likely to break under stress	Machined Aluminum 6061. PLA is not strong enough for this part because it is load-bearing.
10	Hinge board side	PLA	3D printing	Can create more complex 3D shapes and is good for small custom parts	Harder to make quick physical changes after printing	PLA. This part needs to be strong enough and has a more complex shape, but it is small enough that print time is not a major issue.
		Wood	Laser cutting	Faster to produce for flat shapes	Weaker, requires more manual labor, and is limited to simpler flat geometry	PLA. Wood would be faster, but PLA works better because the part is small and may require more complex geometry.
11	C Clamp	PLA	3D printning	Fast to make, cheap, and easy to modify during prototyping	Weaker hold, risk's weaking after repeated use	PLA. 3D printed clamps are easier to manufacture and provide the necessary structural integrity for our prototype.
		Aluminum	CNC	Stronger hold, Sleek look, more durable	Heavier	

Appendix B: Fits and Tolerances Table

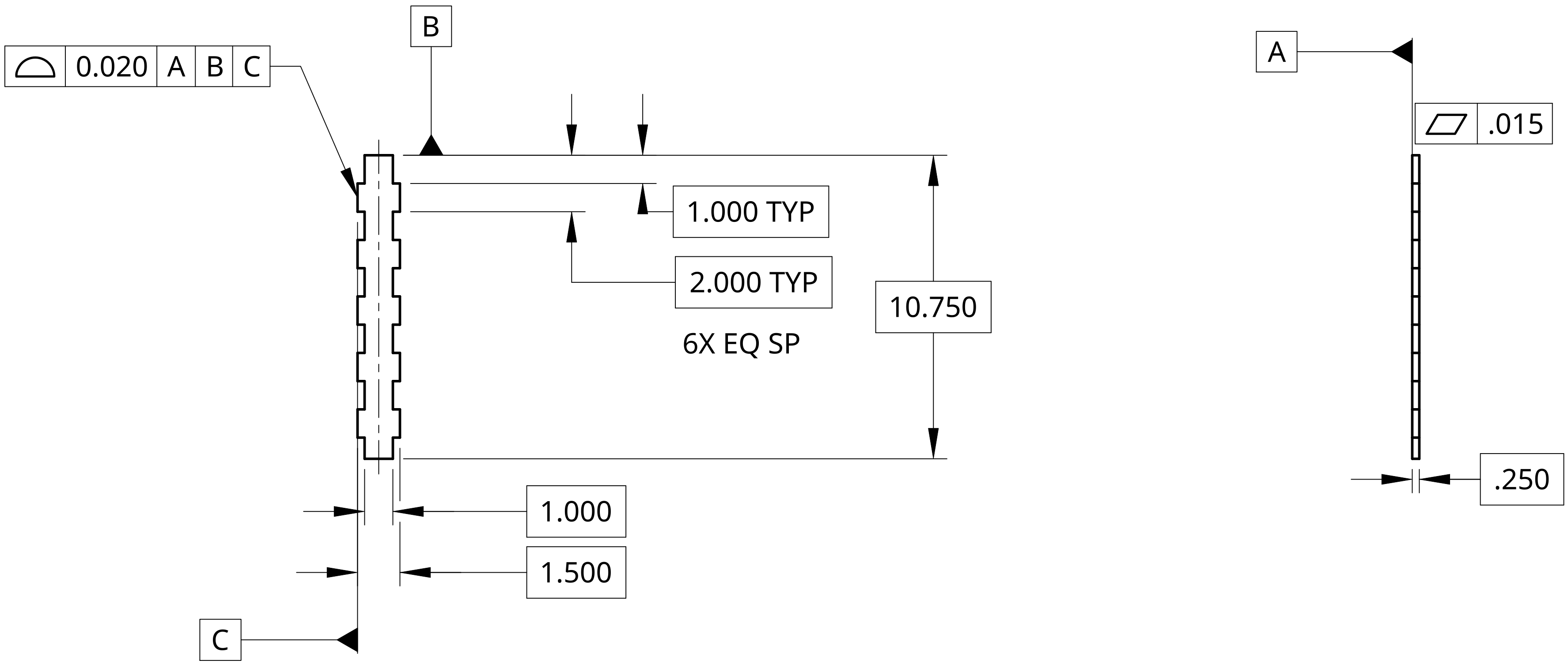
Fit #	Connects component #... (A)	...to component # or external object (B)	Function of fit (e.g. rotation joint, hold phone in place, etc)	ANSI grade of fit (e.g. RC6) or "snap fit"	Component A critical dimension tolerance (e.g. diameter 10.0±0.1 mm)	Component B critical dimension tolerance	Can your chosen manufacturing process deliver the required tolerance(s)?
1	LaptopHolderBoardTop	LaptopHolderBoardSide	Finger joints	LC6	Each finger length: 1 inch + 0.002 inch	Each finger gap length: 1 inch - 0.0008 to - 0.002 inch	No, but with glue, should reduce tolerance and add adhesion
2	HingeMountBoardSide	LaptopHolderBoardSide	Hold the hinge mount in place	LC6	Mount length: 2 inch + 0.002 inch	Mount gap length: 2 inch - 0.0008 to - 0.002 inch	3DP: No, will account for this in the CAD by dimensioning slightly smaller/bigger
3	Hinge	HingeStandoff	Loose rotation joint	RC5	Hinge hole diameter: 6 mm + 0.0007 inch	Standoff diameter: 6 mm - 0.0008 to - 0.0013 inch	3DP: No, will account for this in the CAD, Aluminum standoff: Yes
4	ClampPivotBar	Standoff or Screw	Rotation joint	RC4	Hole diameter: .25 inch + 0.0009 inch	Rod/Screw diameter: .25 inch - 0.0005 to - 0.0011 inch	No, will adjust hole geometry in CAD
5	HingeMountBoardSide	Hinge	Allows for Hinge to pivot	RC7	Mount inner length: 38.1 mm + 0.0025 inch	Hinge length: 38.1 mm - 0.003 to - 0.0046 inch	3DP: No, will acount for this in the CAD
6	LocknutRetainer	Locknut	Hexagonal fit for locknut to prevent nut rotation	LC6	Hex flat-to-flat dist: 5.7 mm + 0.0012 inch	Hex flat-to-flat: 5.7 mm - 0.0004 to - 0.0011 inch	3DP: No, will acount for this in the CAD
7	ScrewCapToScrew	Clamp Screw	Clamping Surface For Screw	LC3	Hole diameter: .25 inch + 0.001 inch	Screw Diameter: 0.25 inch - 0.0005 to 0.001 inch	3DP: No, will account for this in the CAD
8	All screws	All holes	Clearance holes	ASME B18.2.8 Standards in SolidWorks HoleWizard automatic hole sizing			



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ± .05 ANGULAR = ±1 ° .XXX = ± .005 FRACTIONAL = $\pm\frac{1}{32}$	TITLE PART 1 - BOTTOM PANEL
	MATERIAL PLYWOOD
SCALE 1:6	PROCESS LASER CUTTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE TABS AND SAND BURR



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ± .05 ANGULAR = ±1 ° .XXX = ± .005 FRACTIONAL = ±1/32	TITLE PART 2 - TOP PANEL MATERIAL PLYWOOD
SCALE 1:6	PROCESS LASER CUTTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE TABS AND SAND BURR



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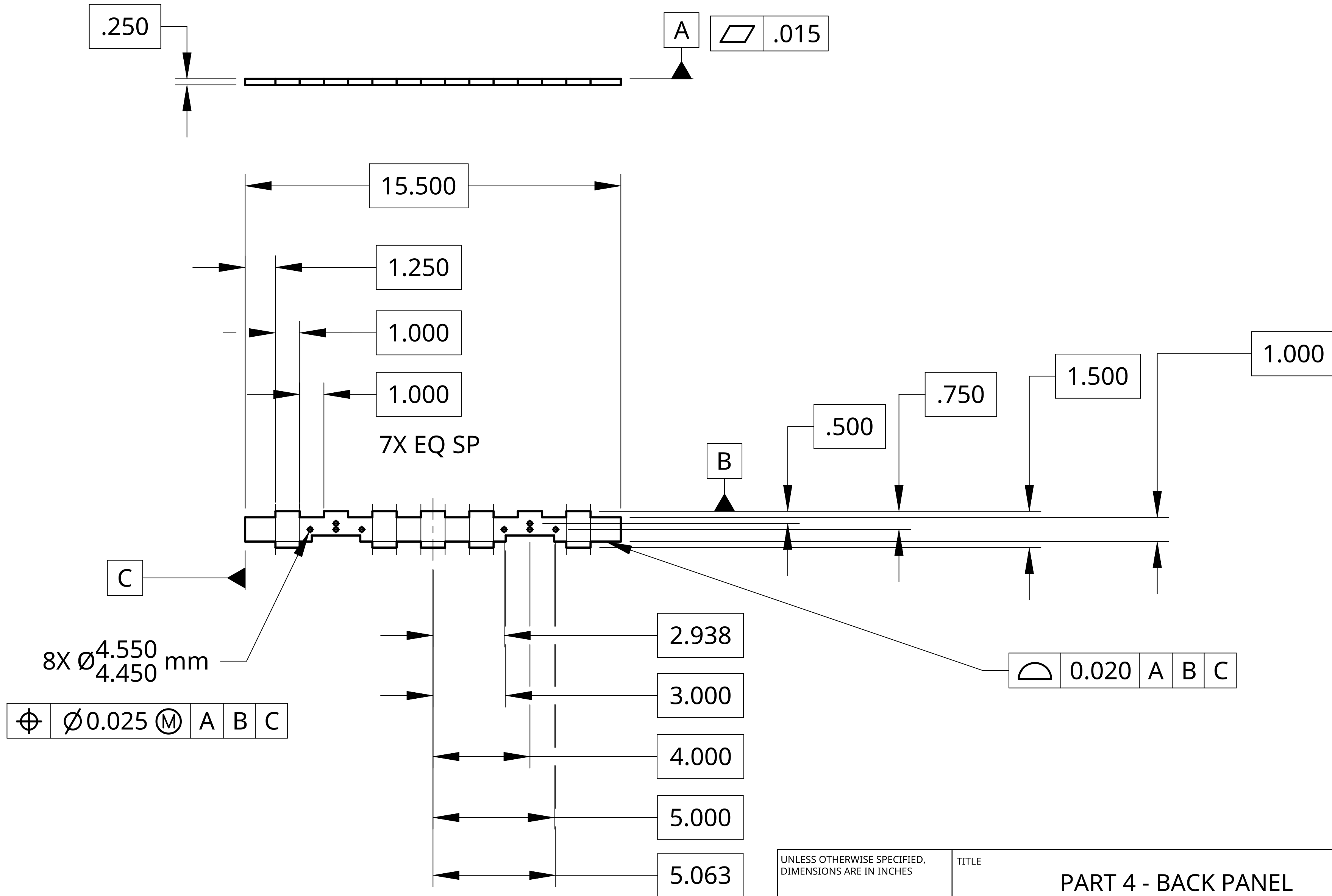
SCALE 1:6	PROCESS LASER CUTTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE TABS AND SAND BURR



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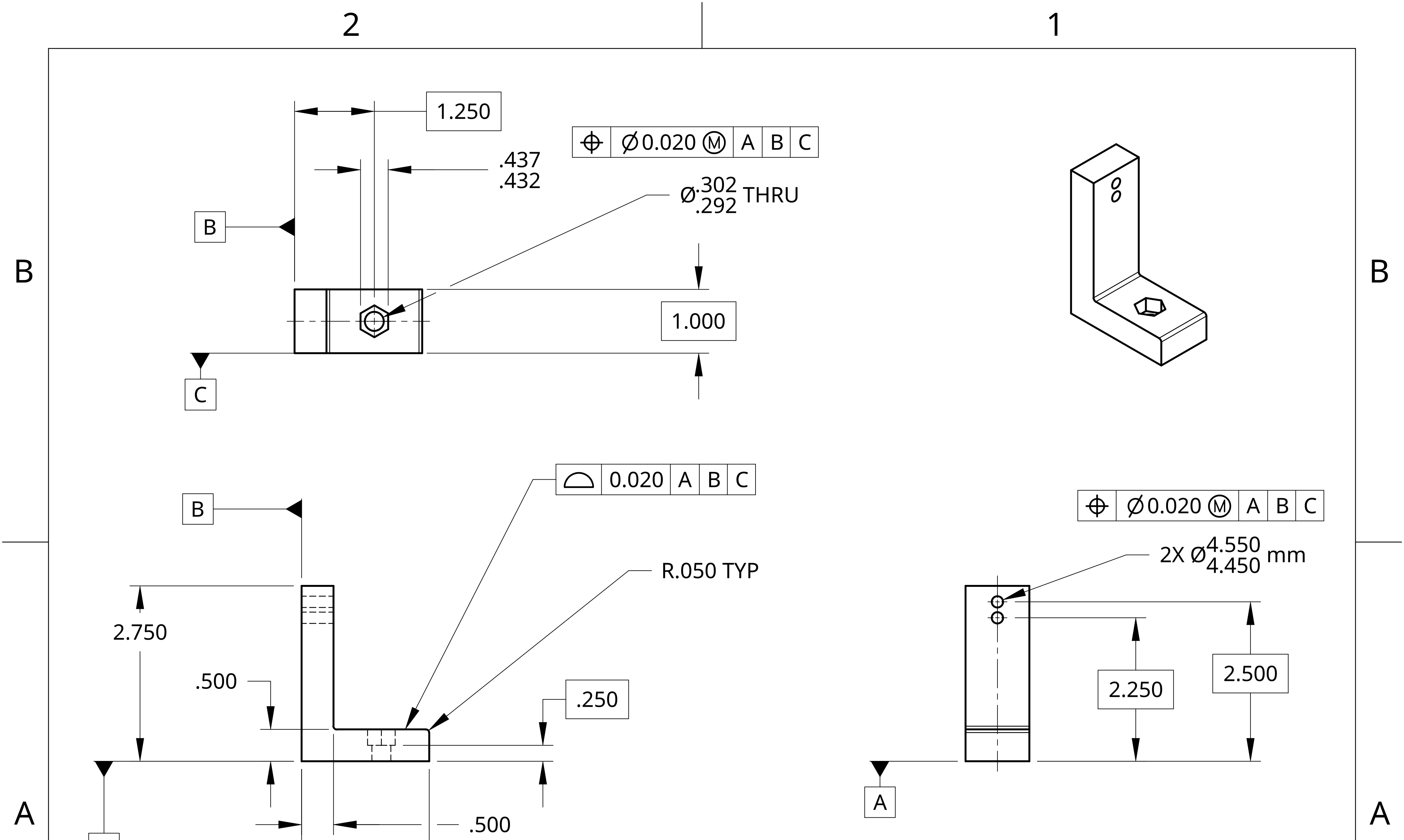
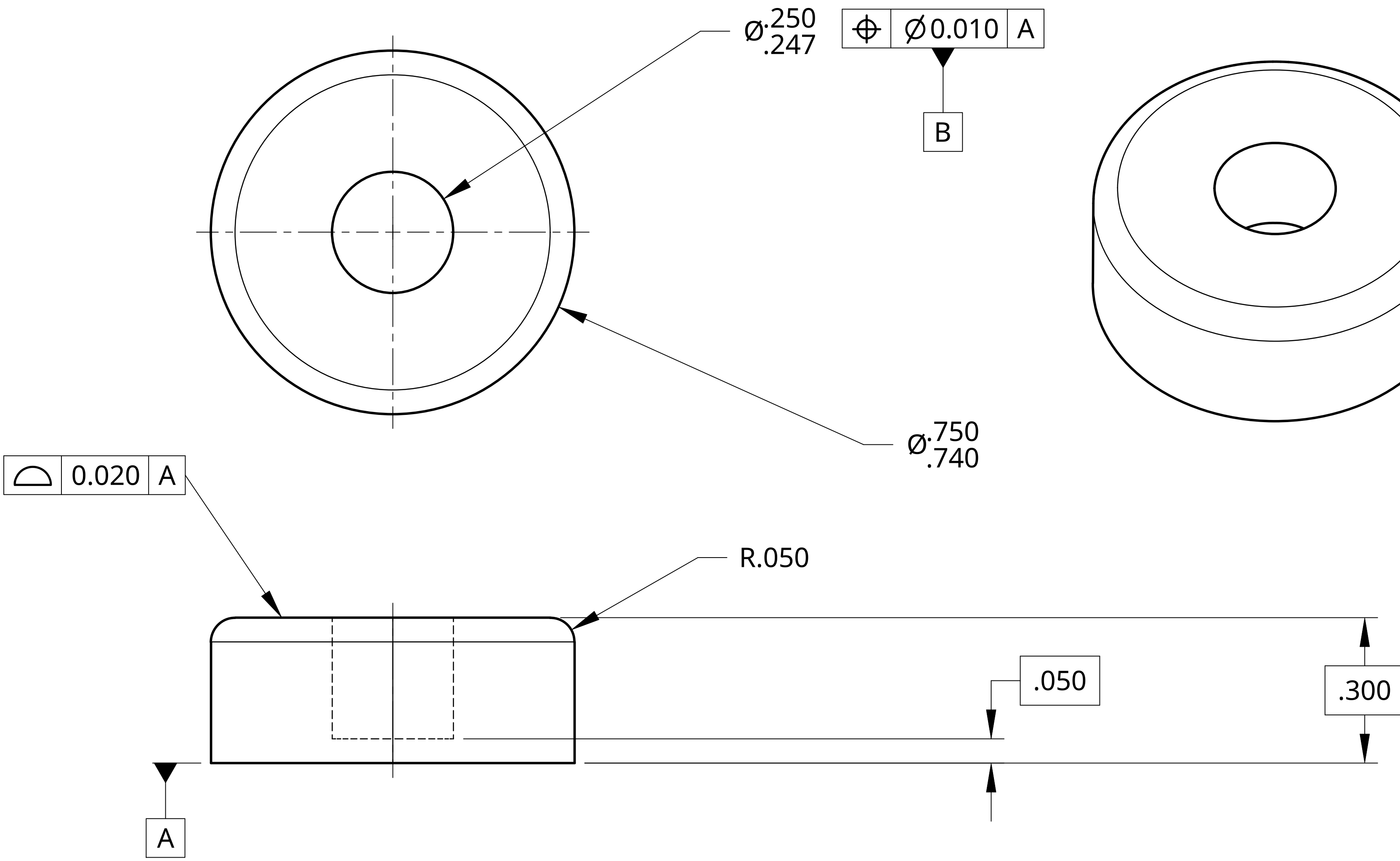
SCALE 1:6	PROCESS LASER CUTTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE TABS AND SAND BURR



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ± .05 ANGULAR = ±1 ° .XXX = ± .005 FRACTIONAL = ±1/32	TITLE PART 5 - L BRACKET
	MATERIAL PLA
SCALE 1:2	PROCESS FDM 3D PRINTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE SUPPORT MATERIAL AND DEBURR

2



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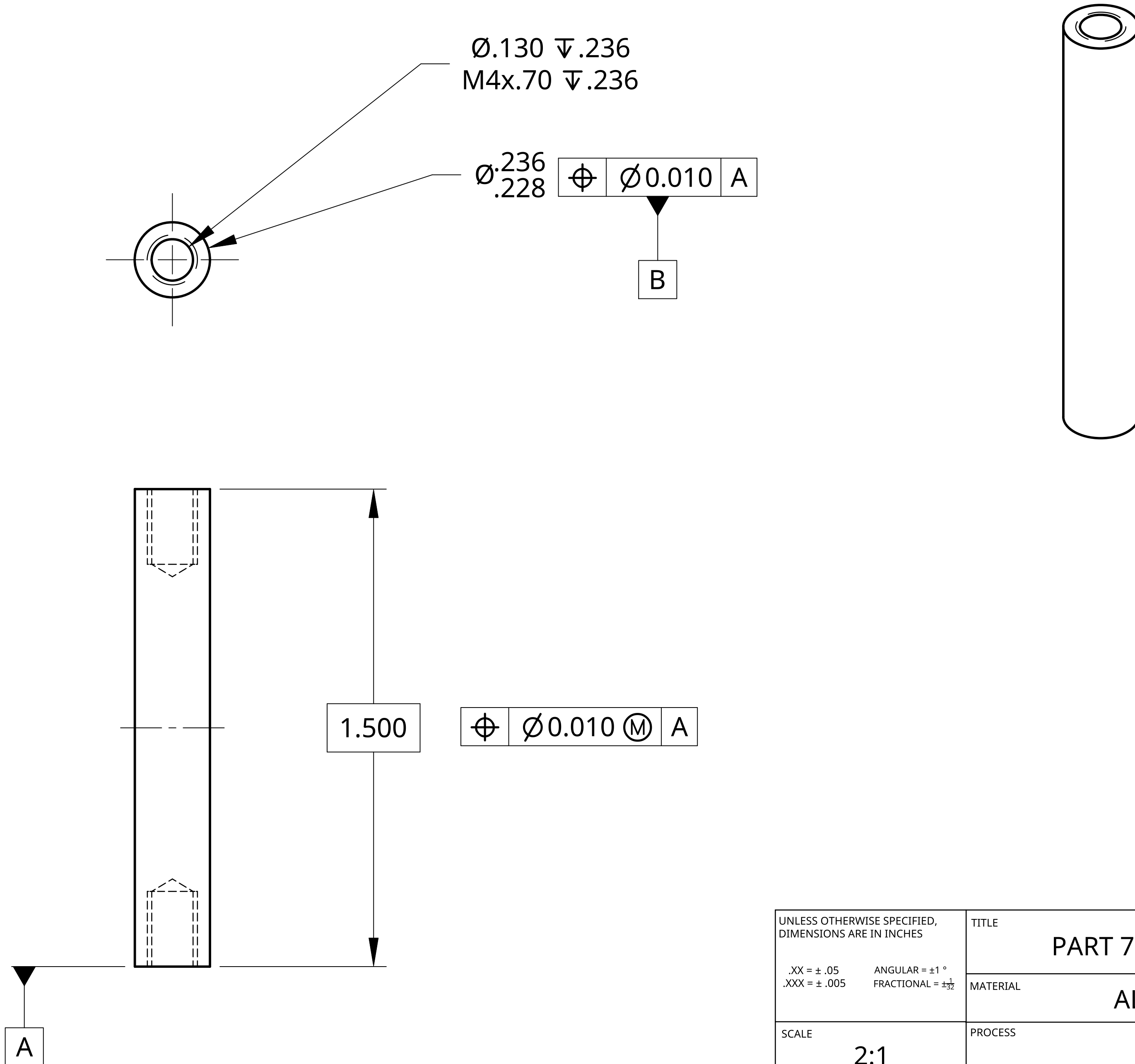
	MATERIAL PLA
SCALE 3:1	PROCESS FDM 3D PRINTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE SUPPORT MATERIAL AND DEBURR



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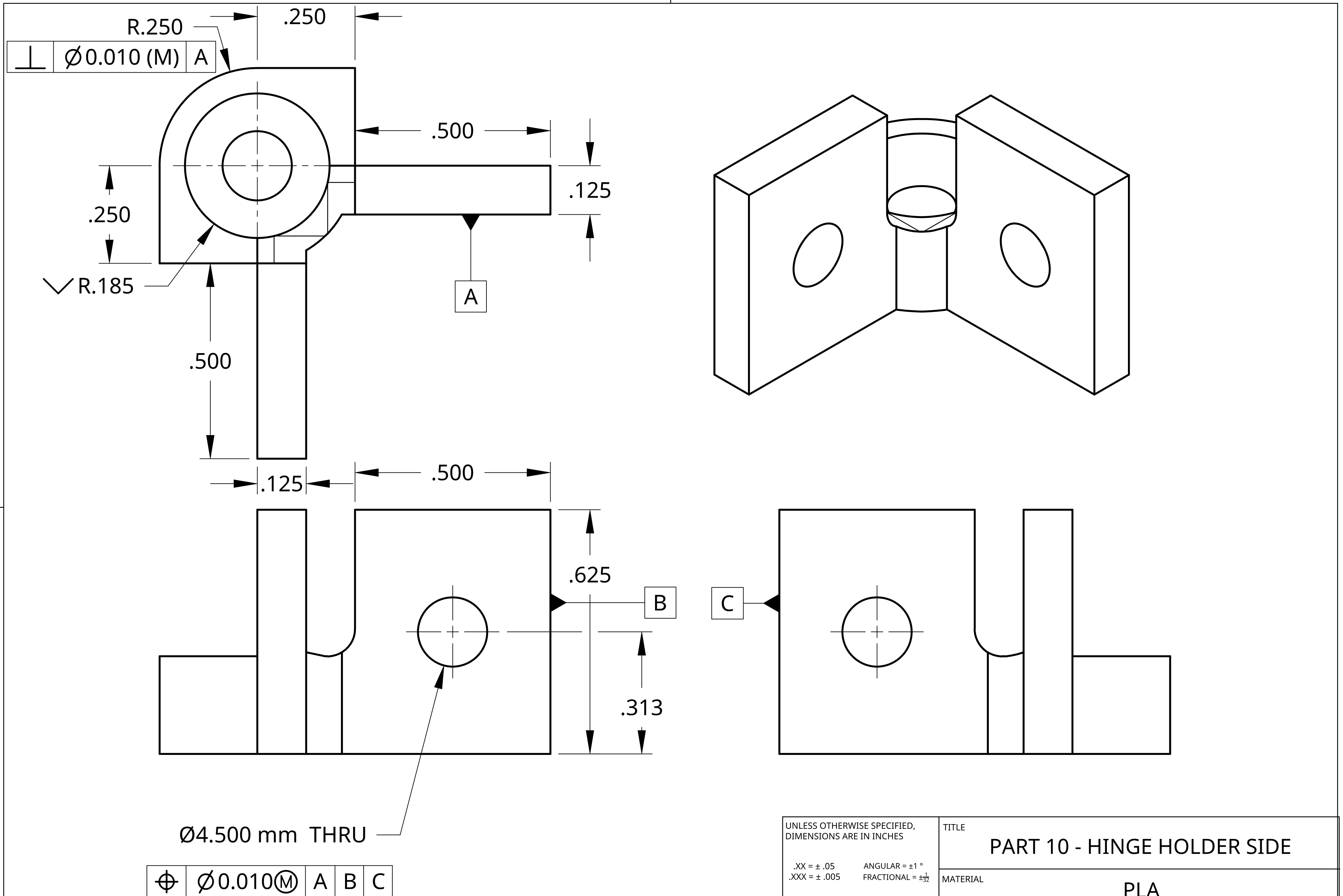
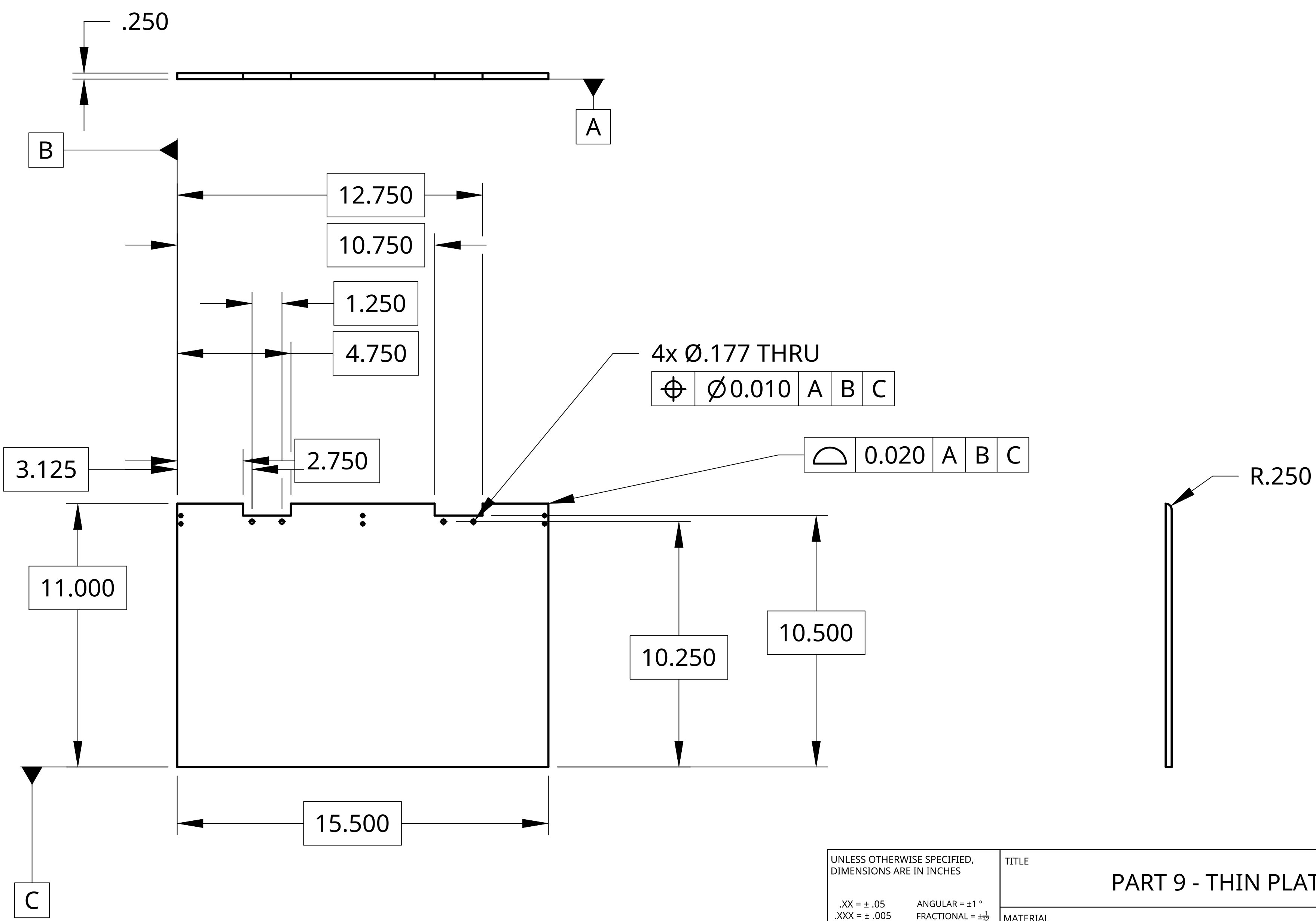
	MATERIAL ALUMINUM 6061
SCALE 2:1	PROCESS MACHINING
THIRD ANGLE PROJECTION	COMMENTS DEBURR AND BREAK SHARP EDGES



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES	TITLE
.XX = ± .05 / .XXX = ± .005 / ANGULAR = ±1 ° / FRACTIONAL = ±1/32	PART 10 - HINGE HOLDER SIDE
	MATERIAL: PLA
SCALE: 3:1	PROCESS: FDM 3D Printing
THIRD ANGLE PROJECTION	COMMENTS: REMOVE SUPPORT MATERIAL AND DEBURR



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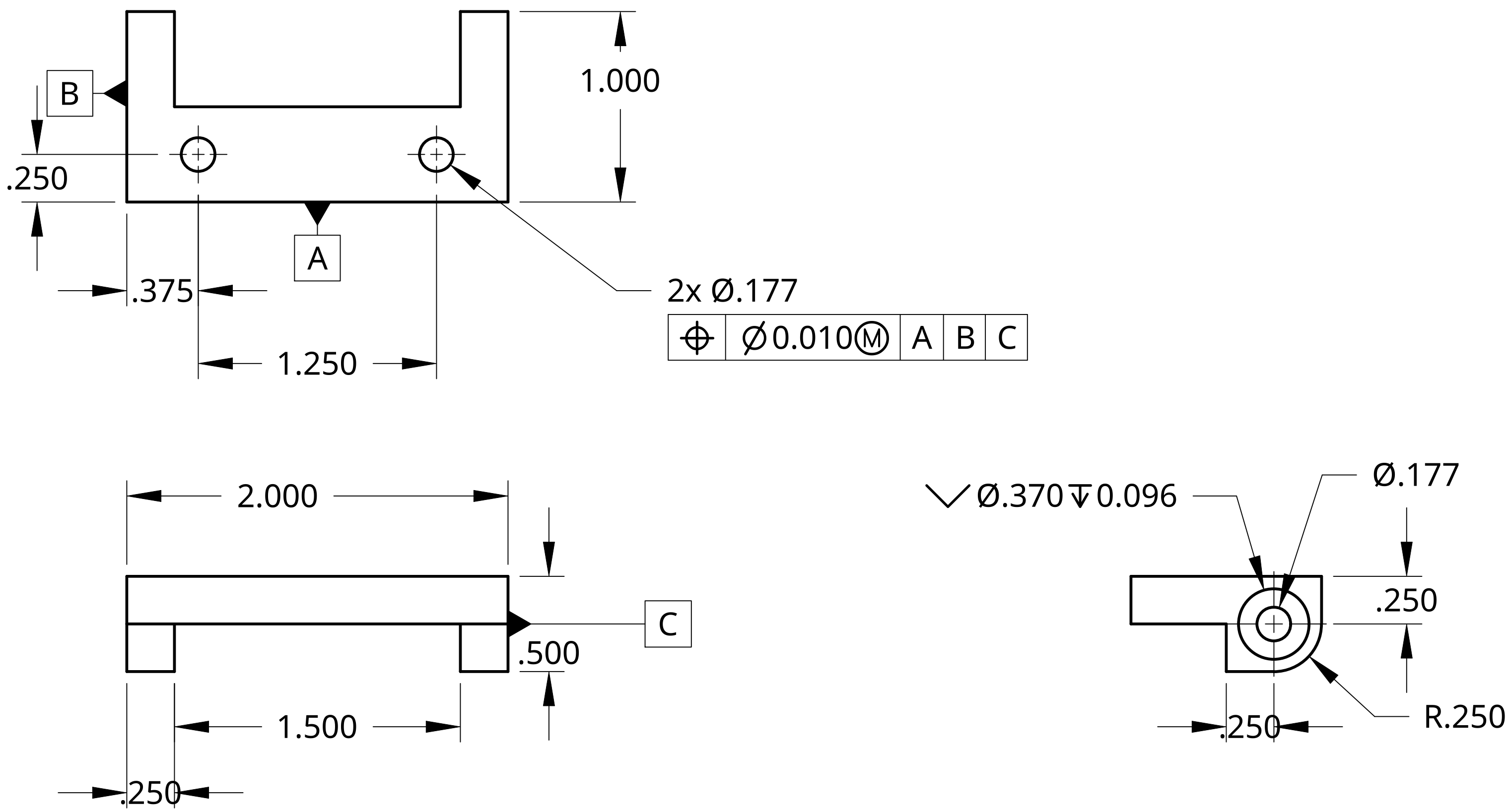
SCALE 1:6	PROCESS LASER CUTTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE TABS AND SAND BURR



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES

.XX = ± .05
.XXX = ± .005

ANGULAR = ±1 °
FRACTIONAL = ±1/32

SCALE

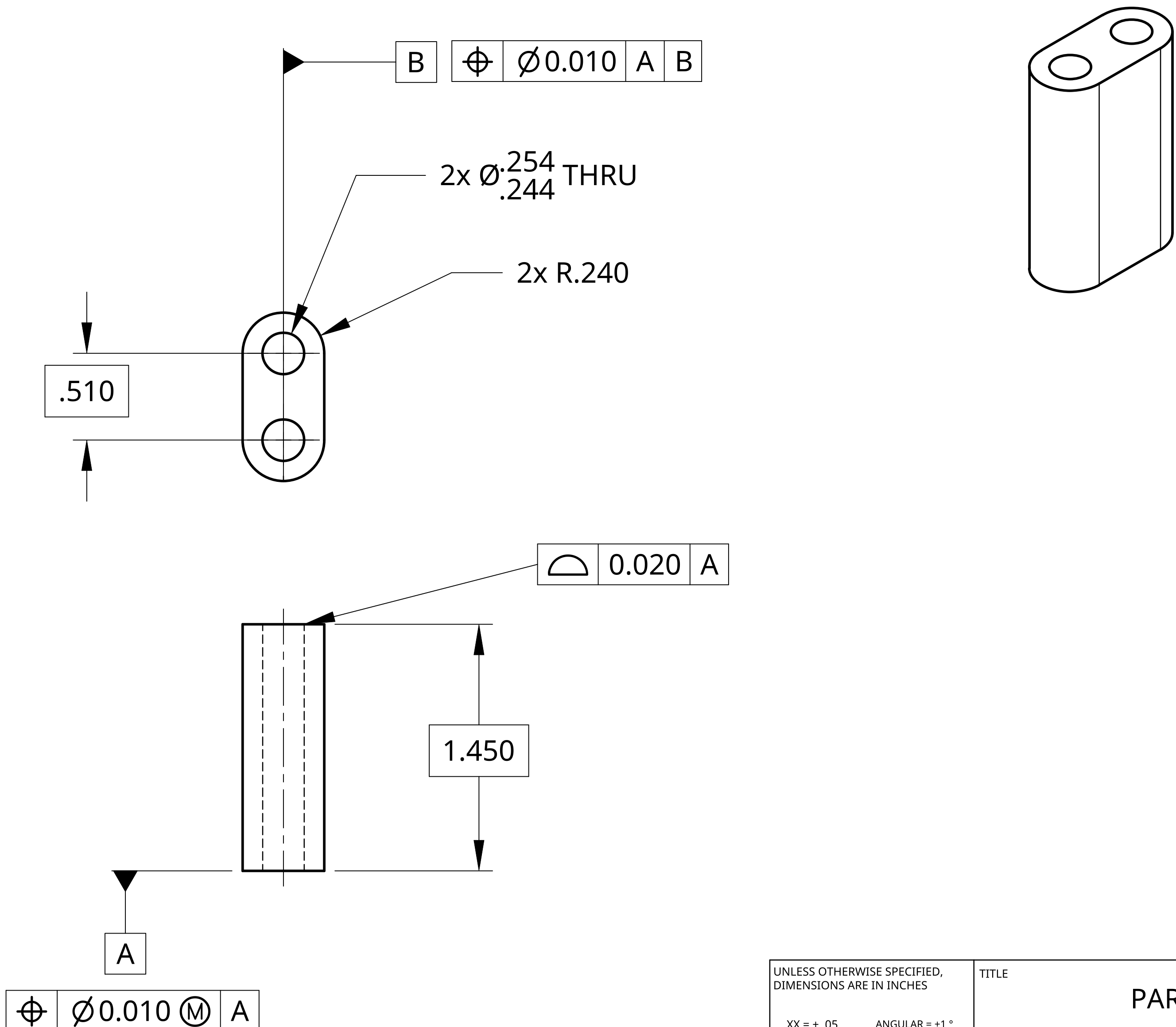
THIRD ANGLE PROJECTION

TITLE
PART 8 - HINGE HOLDER

MATERIAL
PLA

PROCESS
FDM 3D PRINTING

COMMENTS
GLOBAL TOLERANCE: 0.010



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ± .05 ANGULAR = ±1 ° .XXX = ± .005 FRACTIONAL = ±$\frac{1}{32}$	TITLE PART 11 - HINGE
	MATERIAL PLA
SCALE 1:1	PROCESS FDM 3D PRINTING
THIRD ANGLE PROJECTION	COMMENTS REMOVE SUPPORT MATERIAL AND DEBURR